

Return Co-movement

MSCF Investments

Fall 2021 Mini 1

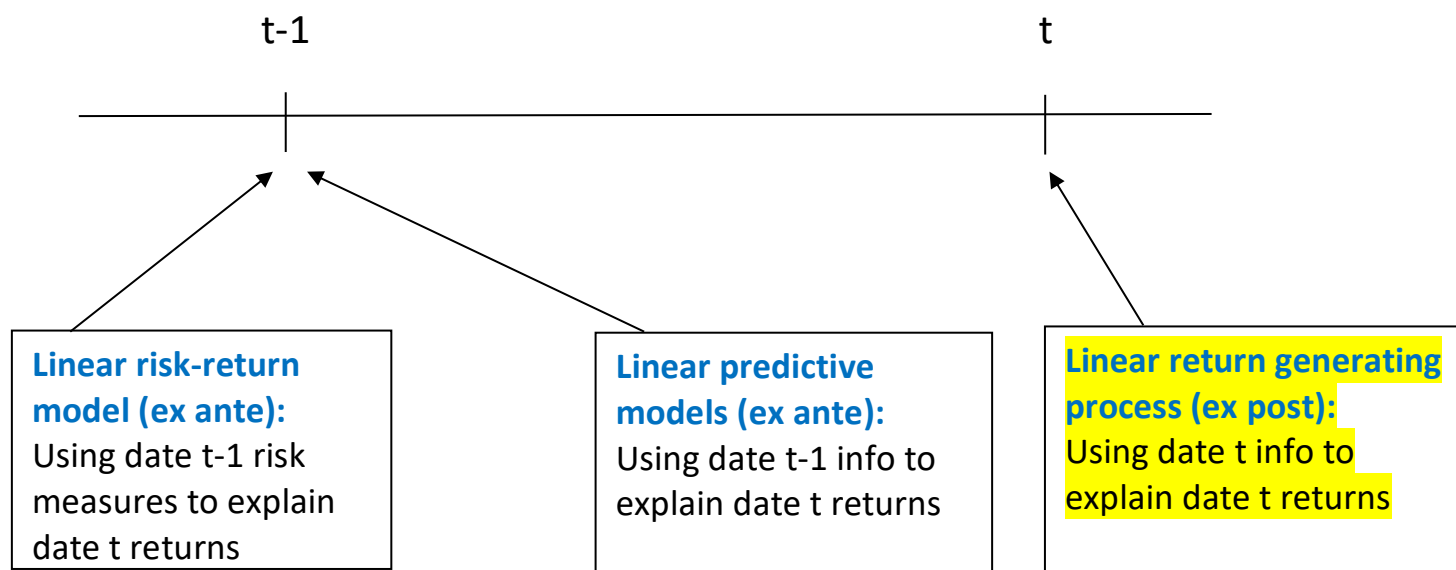
- Linear factor models
- Decompositions: Systematic and idiosyncratic returns
- Fama-French and other examples

Motivation

- **Question:** Why are prices/returns of different securities cross-sectionally correlated?
- Return co-movement due to co-movement in **cash flows** & **discount rates**
 - Stocks?
 - Sovereign bonds?
 - Corporate bonds?
 - Different commodities?
 - Different currencies?

Linear Models

- 3 possible timings relative to date t returns:



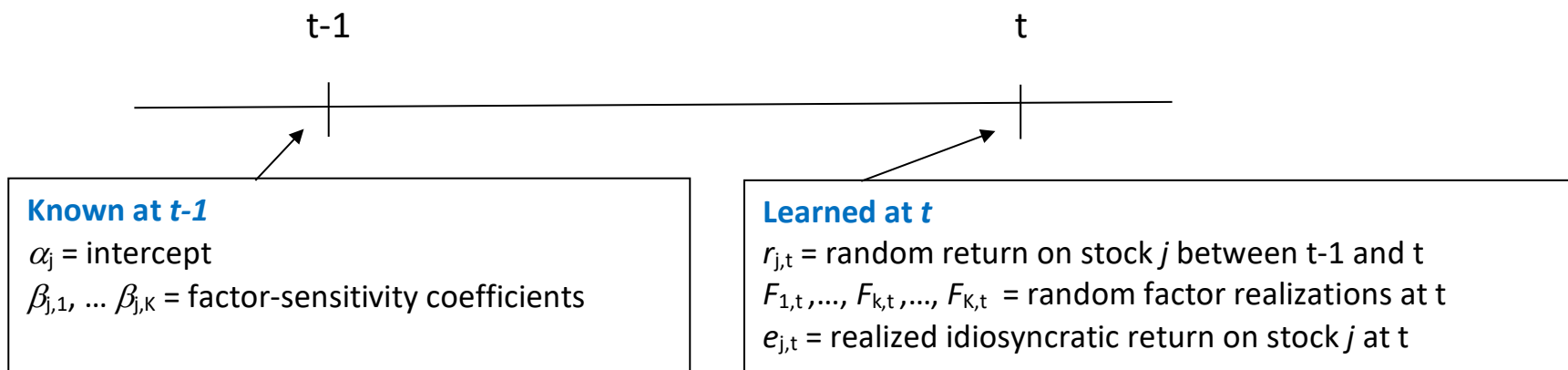
Return-Generating Process: Linear Factor Models

- Individual security returns

$$r_{j,t} = [\text{systematic/macro components}] + [\text{idiosyncratic component}]$$

$$= \alpha_j + \beta_{j,1} F_{1,t} + \dots + \beta_{j,k} F_{k,t} + \dots + \beta_{j,K} F_{K,t} + e_{j,t}$$

- **Systematic factors** $\{ F_{1,t}, \dots, F_{k,t}, \dots, F_{K,t} \}$ affect many different securities. Sources of common randomness across stocks. No j subscripts on F_{kt} 's!
- **Factor sensitivities** $\{ \beta_{j,1}, \dots, \beta_{j,K} \}$ are different for different securities j because different industries are affected differently by factors. Security j subscripts on coefficient $\beta_{k,j}$ for factor k .
- **Residual return** $e_{j,t}$ is specific to security j if factors capture all comovement
- Ex post **return attribution, not predictive**: Use things learned at t that explain returns at t .



Return co-movement

□ Covariance in linear factor model

- Consider two securities j and i
- Suppose residual returns are idiosyncratic in that
 - Residuals are uncorrelated with factors: $\text{Cov}(e_{j,t}, F_{k,t}) = \text{Cov}(e_{j,t}, F_{k,t}) = 0$ for j, i, and all k
 - Residuals are uncorrelated across different securities: $\text{Cov}(e_{j,t}, e_{i,t}) = 0$

$$\begin{aligned}\text{cov}(r_{j,t}, r_{i,t}) &= \text{cov}(\alpha_j + \beta_{j,1} F_{1,t} + \dots + \beta_{j,K} F_{K,t} + e_{j,t}, \alpha_i + \beta_{i,1} F_{1,t} + \dots + \beta_{i,K} F_{K,t} + e_{i,t}) \\ &= \sum_{k=1, \dots, K} \beta_{j,k} \beta_{i,k} \text{var}(F_{k,t}) + \sum_{k=1, \dots, K} \sum_{z \neq k} \beta_{j,k} \beta_{i,z} \text{cov}(F_{k,t}, F_{z,t})\end{aligned}$$

- ### □ **Idea:** Different securities' returns driven by common randomness but different sensitivities.

Estimation

- If a linear factor model is a good model of the return on security j ,

$$r_{j,t} = \alpha_j + \beta_{j,1}F_{1,t} + \cdots + \beta_{j,k}F_{k,t} + \cdots + \beta_{j,K}F_{K,t} + e_{i,t}$$

- Then can estimate using an OLS regression

$$r_{j,t} = \hat{\alpha}_j + \hat{\beta}_{j,1}F_{1,t} + \cdots + \hat{\beta}_{j,k}F_{k,t} + \cdots + \hat{\beta}_{j,K}F_{K,t} + \hat{e}_{i,t}$$

Types of systematic factors

Macroeconomic variables observed at date t :

- % change in GDP, industrial production, etc. Change in inflation rate (e.g., CPI), oil prices

Contemporaneous returns on other assets between $t-1$ and t :

- Return on S&P 500, EAFE, or other “market” index proxies
- Returns on “industry” portfolios
- Returns on long-dated treasury bonds, AAA bonds, junk bonds
- Index and bond returns are used to summarize the financial implications of other unidentified “deeper” macroeconomic variables.

Accounting variables:

- Percentage changes in aggregate corporate earnings

Other factors:

- Changes in consumer confidence
- Bond yields and yield spreads, option implied volatilities, FX rates
- Events need not be numerical to have a numerical impact on prices.
- Market “digests” financial implications of non-numerical events.
- Examples: Election results, political events (Brexit), regulatory & legal changes (taxation of overseas corporate cash), wars, natural disasters, weather, court decisions, etc.

Example: Market model

$$r_{j,t} = \alpha_j + \beta_{j,m} r_{m,t} + e_{j,t}$$

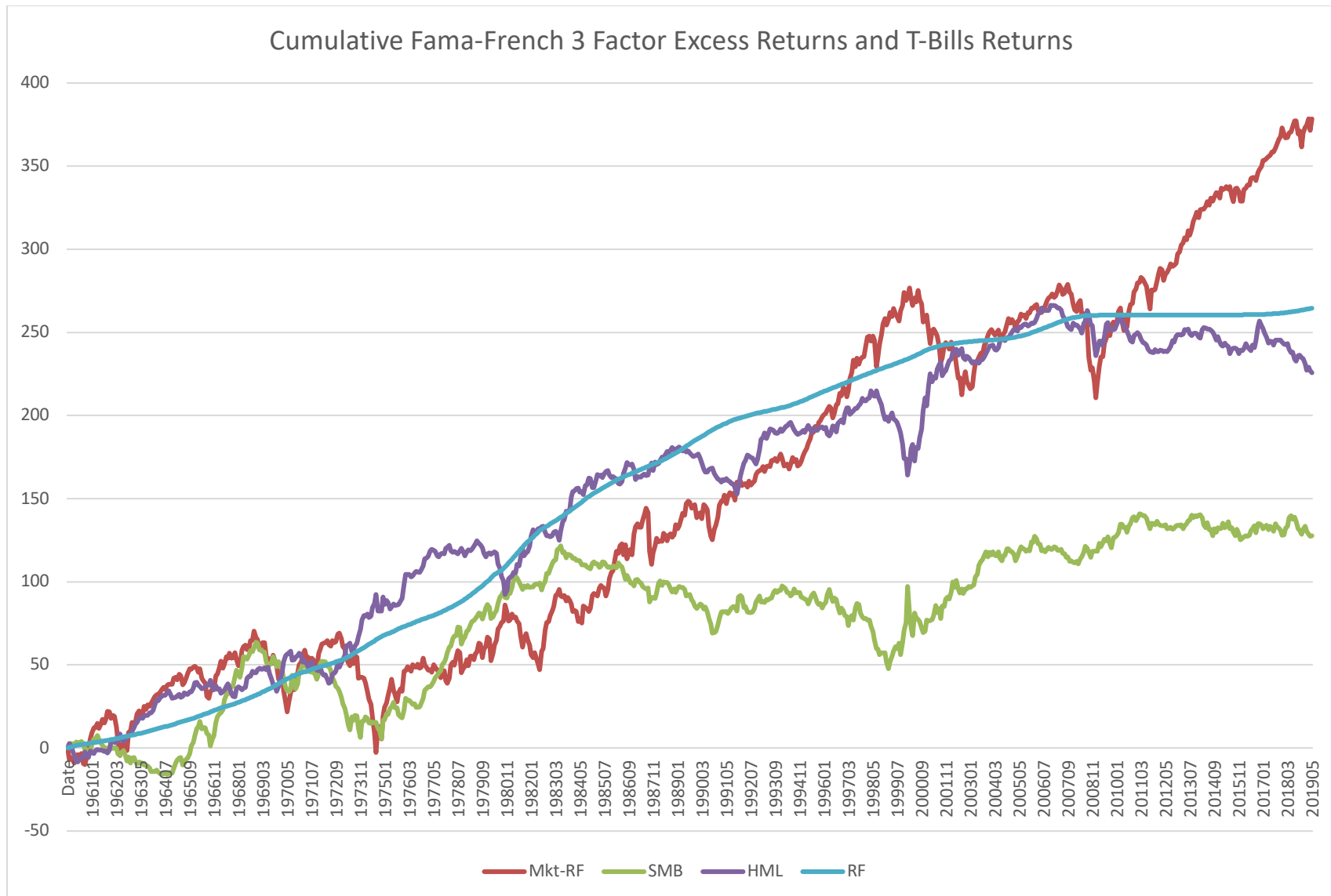
- Only factor is return r_{mt} on broad market index (MSCI World, S&P, EAFA, Wilshire 5000, Nikkei)
- Can use OLS regression to estimate constants α_j and $\beta_{j,m}$ and $SD(e_{j,t})$
- Sometimes mistakenly called “CAPM”. But it’s **NOT** the CAPM. The CAPM is **something else**.
- Interpretation
 - Unconditional expected return: $E[r_{j,t}] = \alpha_j + \beta_{j,m} E[r_{m,t}]$
 - Ex ante conditional expected return: $E_{t-1}[r_{j,t}] = \alpha_j + \beta_{j,m} E_{t-1}[r_{m,t}]$
 - Ex post market component of return: $E[r_{j,t} | r_{m,t}] = \alpha_j + \beta_{j,m} r_{m,t}$
 - Risk decomposition
 - Total variance: $\text{var}[r_{j,t}] = (\beta_{j,m})^2 \text{var}[r_{m,t}] + \text{var}[e_{j,t}]$... **if** $e_{j,t}$ and $r_{m,t}$ are **uncorrelated**
 - Market risk: $SD[r_{m,t}]$
 - Systematic risk of stock j: $(\beta_{j,m})^2 \text{var}[r_{m,t}]$ or sometimes rescaled as $|\beta_{j,m}| SD[r_{m,t}]$
 - Idiosyncratic risk of stock j: $SD[e_{j,t}]$
- Alternative: Often estimated in excess-return form

$$r_{j,t} - r_{f,t} = \alpha_j + \beta_{m,j} [r_{m,t} - r_{ft}] + e_{j,t}$$

Example: 3-factor Fama-French model:

$$r_{j,t} - r_{f,t} = \alpha_j + \beta_{j,m} (r_{m,t} - r_{f,t}) + \beta_{j,HML} HML_t + \beta_{j,SML} SML_t + e_{jt}$$

- $r_{f,t}$ = One-period **risk-free return** (e.g., one-month t-bill if monthly model)
- $r_{j,t} - r_{f,t}$ = **Excess return** on stock j over the risk-free rate
- $r_{m,t} - r_{f,t}$ = **Excess return** on “**market**” portfolio (e.g., S&P 500)
- $HML_t = r_{\text{value},t} - r_{\text{growth},t}$ = Return difference on **long-short** portfolio of **high book-to-market** (value) stocks (**H**) minus the return on a portfolio of **low book-to-market** (growth) stocks (**L**).
- $SML_t = r_{\text{small},t} - r_{\text{large},t}$ = Return difference on long-short portfolio of **small** (low market cap) stocks (**S**) minus the return on a portfolio of **large** (high cap) stocks (**L**).



Example: 5-factor Fama-French model:

$$r_{j,t} - r_{f,t} = \alpha_j + \beta_{j,m} (r_{m,t} - r_{f,t}) + \beta_{j,HML} HML_t + \beta_{j,SML} SML_t + \beta_{j,RMW} RMW_t + \beta_{j,CMA} CMA_t + e_{jt}$$

- Starts with 3-factor Fama-French model and then adds ...
- $RMW_t = r_{\text{robust},t} - r_{\text{weak},t}$ = Return difference for a long-short portfolio that is long robust (high profitability) stocks (**R**) minus the return on a portfolio of weak (low profitability) stocks (**W**).
- $CMA_t = r_{\text{conservative},t} - r_{\text{aggressive},t}$ = Return difference on long-short portfolio of **conservative** (low investment) stocks (**C**) minus return on a portfolio of **aggressive** (high investment) stocks (**A**).

5-Factor Fama-French Portfolio Sorts

- From “A five-factor asset pricing model”, Eugene Fama & Ken French, Journal of Financial Economics, 2015, 1-22.

Table 2

Averages of monthly percent excess returns for value-weight (VW) portfolios formed on (i) Size, B/M, and OP, (ii) Size, B/M, and Inv, and (iii) Size, OP, and Inv, July 1963–December 2013, 606 months.

At the end of June each year t , stocks are allocated to two Size groups (Small and Big) using the NYSE median market cap as breakpoint. Stocks in each Size group are allocated independently to four B/M groups (Low B/M to High B/M for fiscal year $t-1$), four OP groups (Low OP to High OP for fiscal year $t-1$), and four Inv groups (Low Inv to High Inv for fiscal year $t-1$) using NYSE breakpoints specific to the Size group. The table shows averages of monthly returns in excess of the one-month Treasury bill rate on the 32 portfolios formed from each of the three sorts.

	Small				Big			
Panel A: Portfolios formed on Size, B/M, and OP								
B/M →	Low	2	3	High	Low	2	3	High
Low OP	0.03	0.72	0.84	0.93	0.24	0.23	0.37	0.60
2	0.67	0.76	0.88	1.08	0.41	0.50	0.47	0.69
3	0.66	0.88	1.07	1.30	0.40	0.59	0.68	0.91
High OP	0.81	1.13	1.22	1.63	0.53	0.64	0.79	0.71
Panel B: Portfolios formed on Size, B/M and Inv								
B/M →	Low	2	3	High	Low	2	3	High
Low Inv	0.69	0.99	1.18	1.23	0.58	0.70	0.62	0.77
2	0.87	0.92	0.93	1.08	0.49	0.54	0.54	0.60
3	0.84	0.95	1.01	0.97	0.49	0.54	0.56	0.72
High Inv	0.39	0.75	0.87	1.01	0.49	0.44	0.39	0.64
Panel C: Portfolios formed on Size, OP, and Inv								
OP →	Low	2	3	High	Low	2	3	High
Low Inv	0.85	1.01	1.19	1.27	0.63	0.66	0.79	0.70
2	0.94	0.90	0.92	1.04	0.32	0.43	0.64	0.64
3	0.61	0.93	0.94	1.06	0.52	0.57	0.48	0.53
High Inv	-0.09	0.58	0.76	0.76	0.29	0.25	0.38	0.65

Example: Carhart model

$$r_{j,t} - r_{f,t} = \alpha_j + \beta_{j,m}(r_{m,t} - r_{f,t}) + \beta_{j,HML} HML_t + \beta_{j,SML} SML_t + \beta_{j,MO} MO_t + e_{jt}$$

- Starts with the 3-Factor Fama-French factors and adds ...

- MO_t = **Momentum factor** = $r_{winner,t} - r_{loser,t}$

= Return difference in month t between a portfolio of previous “**winner**” stocks (with highest returns over the **preceding** 11 months) and a portfolio of previous “**loser**” stocks (with lowest returns over the **previous** 11 months).

- **Question:** Over/under-reaction? Changing risk?

- Related research

- Jegadeesh (1990): Return reversals over monthly horizons.
- Jegadeesh and Titman (1993): Continuation/momentum over 6-12 month horizons.
- Jegadeesh and Titman (1993): Reversals over multi-year horizons

Example: “Kitchen sink”

$$r_{j,t} - r_{f,t} = \alpha_j + \sum_k \beta_{k,i} (r_{k,t} - r_{f,t}) + \sum_g \beta_{g,i} F_{g,t} + e_{jt}$$

- Investable explanatory variables $r_{k,t} - r_{f,t}$
 - Examples:
 - Excess returns on market portfolio, FF factors, industry portfolios, returns on other asset classes (e.g., bonds)
- Non-investable factors $F_{g,t}$
 - Examples:
 - Difference in returns $r_{\text{Junk},t} - r_{\text{AAA},t}$ for below-investment-grade (junk) bond index vs. AAA debt. Used to capture impact of changing “credit” risk on stocks.
 - Difference in returns $R_{10\text{yr},t} - r_{f,t}$ for 10-year treasuries vs. short-term t-bill rate. Used to capture impact of “term structure” risk on stocks.
 - Macro variables like inflation rate (Inf_t), changes in industrial production.
 - Measures of political stability
- Choice of factors in actual models varies in practice and can be somewhat ad hoc.
- Use “kitchen sink” when goal is simply to maximize **return attribution** (ex post explanatory) power of model (i.e., regression R^2).

Generalization: Time-varying factor-sensitivity

$$r_{j,t} = \alpha_{j,t} + \beta_{j,1,t}F_{1,t} + \dots + \beta_{j,k,t}F_{k,t} + \dots + \beta_{j,K,t}F_{K,t} + e_j,$$

- where coefficients $\beta_{j,k,t}$ change according to some model

$$\alpha_{j,t} = \alpha_j(\text{time } t, \text{ other variables})$$

$$\beta_{j,1,t} = \beta_{j,1}(\text{time } t, \text{ other variables})$$

...

$$\beta_{j,K,t} = \beta_{j,K}(\text{time } t, \text{ other variables})$$

- See research by Wayne Ferson

Question: Do factor sensitivities change over the business cycle? (See HW)

Learning objectives

- Concept: Linear factor models and relation to return co-movement
- Definitions of factors in Fama-French models
- Estimation using OLS regression (see HW)
- Compute corresponding return covariances